

Noise-tailored two-qubit gates on a high-contrast featured dephasing model

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June 12, 2026

This note is the companion to the June 11 report: there, the paper’s pipeline ran on the experimentally anchored Tarucha-class model and the noise-tailoring margin came out a reproducible but modest $1.5\text{--}1.7\times$. Here the same pipeline runs on a featured model — every structural ingredient still a measured type of spectral feature, but their placement and contrast chosen so that the value of the full spectral reconstruction is visible at a glance. Three requirements were imposed in the June 12 discussion: the working infidelities must be realistic for today’s devices; the reconstruction must genuinely capture the model — every harmonic of all six spectra within its stated error bars; and the demonstration must articulate what the two-qubit part of the reconstruction buys. Throughout, the optimizer never sees ground truth, and every quoted “true” infidelity is evaluated exactly on the analytic model. All numbers June 12, 2026: the characterization is one 128k-shot reconstruction (everything downstream — designs, budgets, bands — is built on it), and the SPAM comparison is three 64k arms run on identical noise.

1 The noise model: what we assume, and why

Two correlated local fields, not one shared bath. The skeleton is unchanged from the anchored model: two partially correlated local electric fields e_A, e_B (shared fraction 0.8), the exchange coupling responding to their *difference*, $\zeta_{12} = a e_A - b e_B$ with $b \gtrsim a$, plus independent qubit-local processes. This is the minimal structure behind Yoneda’s measured $(+, +, -)$ coherence sign pattern, and it pins the signs and relative phases of all three cross-spectra rather than leaving them free. Four ingredients are then recomposed on this skeleton, each a measured feature type with its amplitude as the stated modeling choice.

A slow, common-mode carrier holds the coherence budget. A slow bath (correlation time $\sim 400\tau$, in-band ω^{-2} tail) carries the entire $T_2^* = 3500\tau$ per qubit — the measured Ge-hole value ($17.6\mu\text{s}$ at the $\tau = 5\text{ ns}$ anchor), equivalently purified ^{28}Si at $\tau \approx 7\text{ ns}$. Its in-band tail is what defeats the low CDD orders. Structurally, the carrier is *common-mode* between the qubits at fraction $c_{\text{QS}} = 0.85$, with the remainder qubit-local: slow drifts of this class (global field, thermal, common charge motion) hit both qubits together, and inter-qubit coherences up to 0.7 are what Yoneda measured at low frequency. Two consequences are wired into the construction: the carrier appears in $\text{Re } S_{1,2}$ (and only there — a common-mode drift cancels in the difference coupling that drives ζ_{12} , so $S_{12,12}, S_{1,12}, S_{2,12}$ carry none of it), and its *sign* at low frequency is a measurable prediction with design consequences (Sec. 3).

The trap structure: a defect line with harmonics. On the qubit-local channels, a narrow defect line at $\omega_0\tau = 0.051$ with harmonics at $\{2, 3, 4\}\omega_0$ (cf. the ^{73}Ge Larmor line and its measured second harmonic surviving isotopic purification), plus one line at $\omega\tau = 0.372$ covering the top of the band reachable under the control-bandwidth constraint below. The placement is the engineered part: the CDD library’s passbands at $T_G = 320\tau$ fall on the first, second, and fourth harmonic for orders 3/4/5, and because the passbands scale as $1/T_G$ while the lines are fixed, *every doubling of the gate time slides the library down onto the next line*. This is why the margins in Sec. 4 grow with gate time. The lines are strictly *local* defect structure: they appear on $S_{1,1}$ and $S_{2,2}$ only, and the inter-qubit coherence dips at the line frequencies.

A line on the exchange channel only. An independent coupler-defect process (TLF resonance at $\omega\tau = 0.236$ with a weak Connors-type knee) enters ζ_{12} alone. Single-qubit QNS cannot see it; it is structure reserved for the two-qubit spectra, and it is what the idle gate’s ZZ-decoupling has to dodge.

A quiet — but measurable — electrical background. A $1/f^{0.9}$ floor (the measured charge-noise exponent class) roughly 25 dB below the anchored natural-Si level. This is the contrast knob, and it is deliberately placed *at* the reconstruction’s sensitivity rather than below it: between the fourth harmonic and the 0.372 line it leaves the quiet band $\omega\tau \in [0.26, 0.31]$ whose *measured* level, not any line, sets what an informed design can reach.

The scenario parameter. A minimum pulse separation of 8τ (40 ns π -pulse spacing at the anchor) applies

identically to the CDD library, the NT optimizer, and every counterfactual. Without it, separation-limited trains park their passbands above the comb's measured band and the comparison decouples from the reconstruction; with it, every sequence lives on spectra the protocol has measured.

quantity	value	anchor
time unit (min. pulse separation / 8)	$\tau = 5$ ns	Ge-hole / fast-Rabi anchor
T_2^* (both qubits)	$3500 \tau = 17.6 \mu\text{s}$	measured Ge-hole T_2^*
Gaussian FID $T_2 = 2/S_{a,a}(0)$	$3.10 \times 10^3 \tau$	model output
slow-carrier correlation time / share	$400 \tau / c_{\text{QS}} = 0.85$ common-mode	Yoneda low- f coherence class
defect line + harmonics	$\omega_0 \tau = 0.051, \{2, 3, 4\} \omega_0, 0.372$	^{73}Ge -line / trap family
coupler (ZZ) line + knee	$\omega \tau = 0.236$; knee at 0.025	shared-TLF class, stylized
electrical floor	$1/f^{0.9}, -25$ dB vs anchored	charge-noise exponent class
quiet band (between lines)	$\omega \tau \in [0.26, 0.31]$	set by the engineered lines
control bandwidth	min. pulse separation $8\tau = 40$ ns	scenario parameter
gate windows studied	CZ: $T_G = 40$ – 2560τ ; idle: 320 – 10240τ	0.01 – $3.3 T_2^*$

Table 1: The dephasing scale and where the gates sit relative to it.

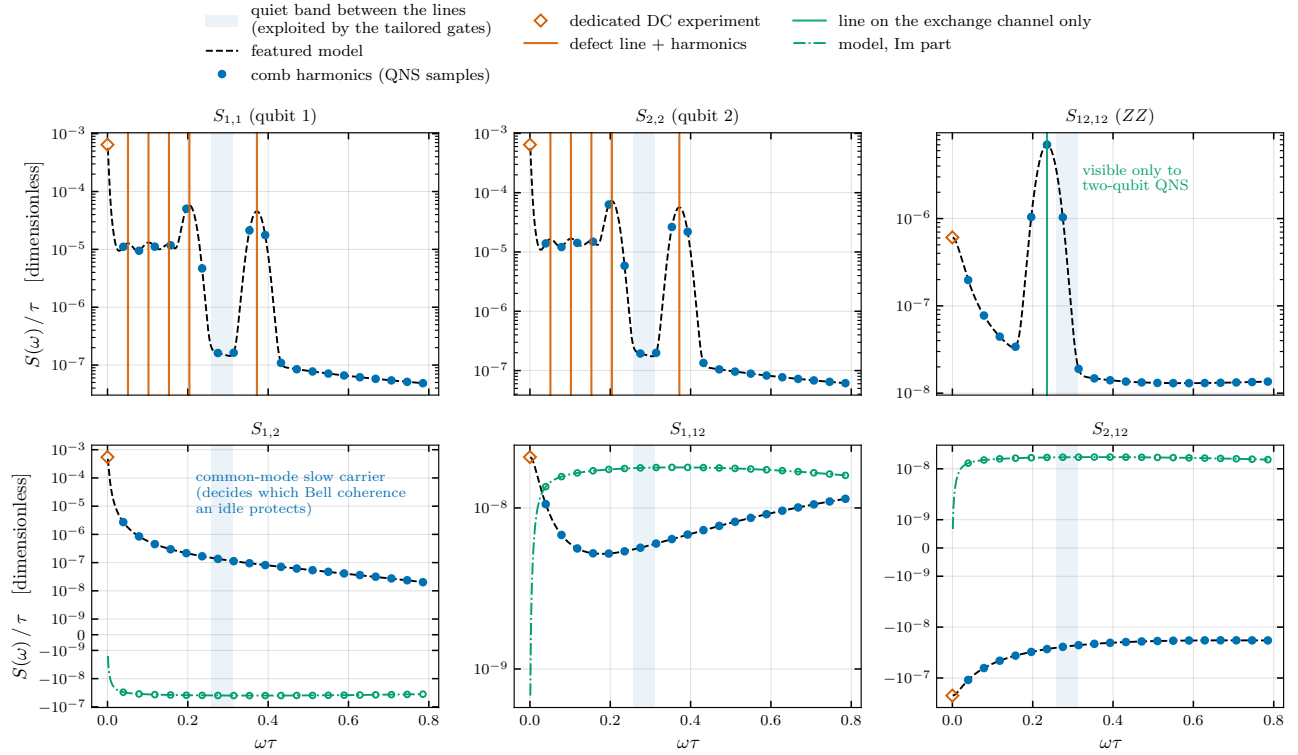


Figure 1: The featured two-qubit dephasing model. Each panel is one of the six spectra the QNS protocol reconstructs: dashed black is the analytic model (top row: the two single-qubit channels and the ZZ channel; bottom row: the three cross-spectra, real parts in black, imaginary parts in green). Blue dots mark the 20 comb harmonics $\omega_k \tau = 2\pi k/160$ the protocol samples; open diamonds are the dedicated DC experiments. Vertical orange lines: the defect line and its harmonics, which live on the qubit-local channels only; the coupler resonance (green) lives on $S_{12,12}$ *only* — structure invisible to single-qubit QNS. The common-mode slow carrier dominates $\text{Re } S_{1,2}$ at low frequency (its low- f rise in the $S_{1,2}$ panel); which Bell coherence an idle protects depends on its measured sign. Shaded band: the quiet window between the lines that the tailored designs exploit.

2 Characterization: the blind reconstruction recovers the truth

The protocol is identical to the anchored study — the same comb, the same estimators, the same DC experiments — run once at the budget a one-time device characterization affords: 128k shots per estimator, with the harmonic sweeps extended to 2560τ ($0.73 T_2^*$; about fifteen minutes of simulation wall time). The acceptance standard is the one set in the June 12 discussion: *within error bars everywhere, with bars that inform*. Table 2 gives the result against the analytic truth, per spectrum.

spectrum	median rel. dev.	median pull	max pull	within 2σ
$S_{1,1}$	1.8%	0.35	1.80	100%
$S_{2,2}$	0.8%	0.34	2.07	95%
$S_{12,12}$	5.1%	0.79	2.20	85%
$S_{1,2}$	6.6%	0.43	1.96	100%
$S_{1,12}$	16.1%	0.67	2.17	95%
$S_{2,12}$	6.6%	0.54	1.81	100%

Table 2: Blind 128k reconstruction vs. truth (pull = deviation / quoted total uncertainty; 20 harmonics per channel). Every tooth of every channel sits within 2.2σ of truth — across the 120 teeth, the handful in the $2\text{--}3\sigma$ band is what correct coverage predicts. The bars inform: median signal-to-bar ratios run 4–25 per channel, and every engineered feature (the defect harmonics on $S_{1,1}/S_{2,2}$, the coupler line on $S_{12,12}$, the common-mode carrier in $\text{Re } S_{1,2}$) is detected at 5σ or better. The double-echo DC experiment lands $S_{12,12}(0) = 6.19 \times 10^{-7}$ against a true 6.07×10^{-7} .

Two honest footnotes to “tight everywhere”. First, harmonics adjacent to the lines on the smallest channels carry locally widened statistical bars: the probe sequences decohere faster while sweeping across a line, a measurement back-action that no shot budget removes; those teeth remain within bars. Second, the slope-fit DC experiments on $S_{1,1}/S_{2,2}$ cannot see the full quasistatic carrier and are flagged *undetermined* by the protocol itself (the stored values are first-harmonic floors); the double-echo $S_{12,12}(0)$ experiment, by contrast, is determined and lands within 2% of truth. The flags propagate to the gate stage, which treats those DC points accordingly — part of the protocol, not a patch.

SPAM. The three-arm comparison (Fig. 2) reruns the full experiment suite with hardware-realistic SPAM injected ($\alpha_M \sim 0.95\text{--}0.97$, $\alpha_{SP}^z \sim 0.98\text{--}0.99$, nonzero readout asymmetries and transverse preparation errors), at 64k shots per arm on *identical noise* (record/replay), so the arms differ only in what SPAM does and what mitigation recovers. The result reproduces the anchored study’s conclusion on the featured landscape: the mitigated arm is statistically indistinguishable from the SPAM-free reference (maximum deviation 0.02σ across all six spectra), while the raw arm is biased at up to 3.2σ per channel — modest in σ ’s only because this quiet landscape’s bars are wide. In *relative* terms the corruption is exactly where a tailored design leans: up to 86–106% on the self-spectra (the carrier head at the first harmonic, a line shoulder), and up to $\sim 9\times$ on the quiet ZZ teeth whose measured level sets the ZZ error floor. The SPAM-robust protocol remains immune by construction at its stated price (it cannot reconstruct $S_{1,12}, S_{2,12}$); whether that price matters is a gate-level question, answered next.

3 What the gate design needs from the spectra, and what the physics does

With a reconstruction that captures the landscape, the question is how much of it the control design actually consumes — and what the parts the design does *not* consume are for. We answer with the same two experiments as the anchored study, run for both gates (Fig. 3), plus the demonstration the cross-spectra exist for (Fig. 4).

Spectral knowledge. We re-ran the blind NT optimization while progressively stripping channels from the optimizer: the two single-qubit spectra alone $\rightarrow$ the three self-spectra $\rightarrow$ the SPAM-robust protocol’s four $\rightarrow$ all six (the entangling gate at $T_G = 320\tau$; the idle at 640τ and 2560τ , best over the repetition number M). The structure of the result differs by gate, and the difference is the physics. For the *entangling* gate the ladder is flat: single-qubit knowledge alone lands within 7% of the full design (3.94 vs 3.69×10^{-4}), and the three-self,

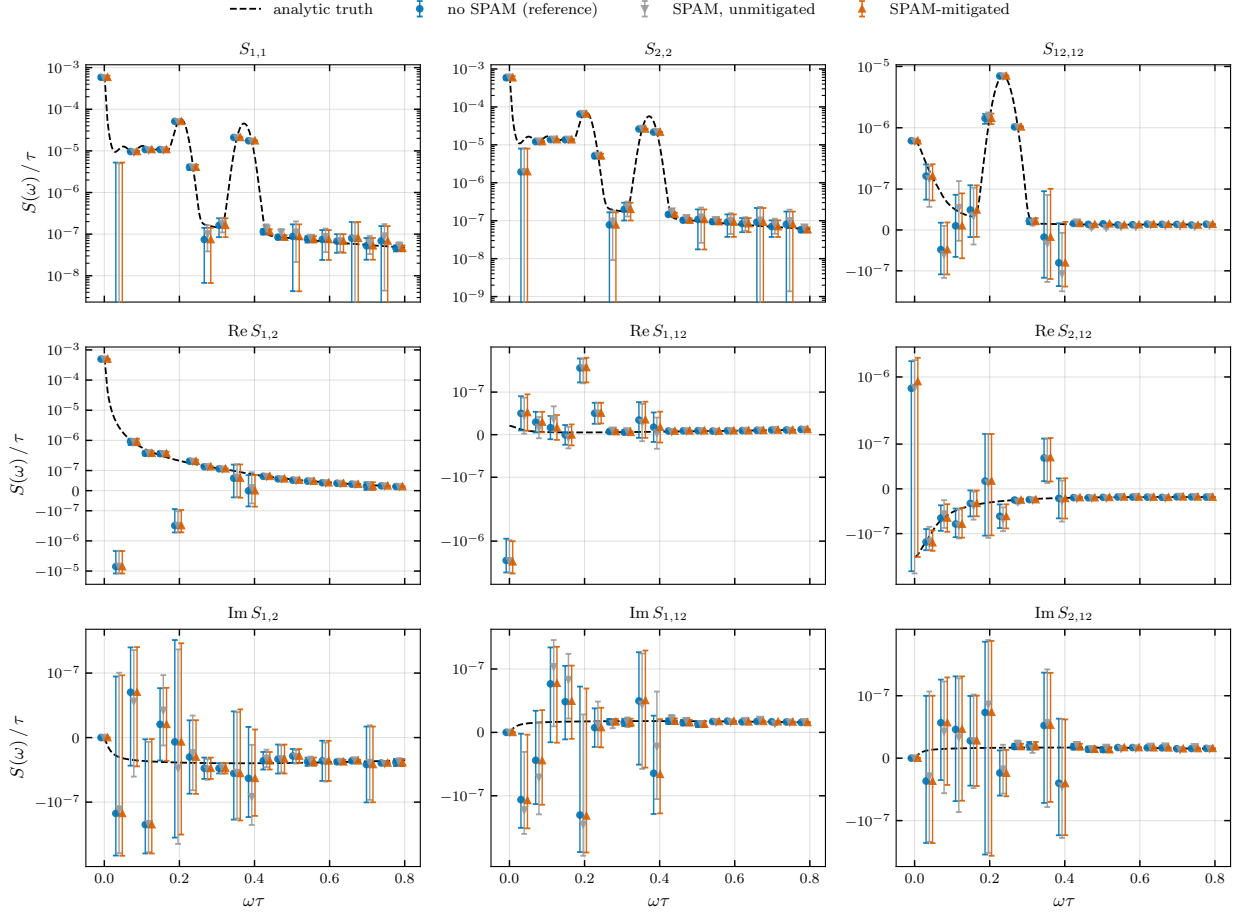


Figure 2: The three reconstruction arms at 64k shots on identical noise, against analytic truth (dashed): reference (no SPAM, blue circles), raw (SPAM, unmitigated, grey triangles), SPAM-mitigated (orange). Bars are the quoted total (statistical + systematic) uncertainties; arms are slightly offset in ω for legibility. Top row: the three self-spectra; middle/bottom rows: real and imaginary parts of the cross-spectra. The raw arm’s bias is largest on the line heights; the mitigated arm sits on the reference everywhere.

robust-four, and six-spectrum designs are the *identical gate* — the same NT(30,29) winner at the same true infidelity. The CZ’s near-synchronous trains put almost no filter weight on the coupler line, and the phase constraint pins the y_{12} filter’s DC weight regardless of what the optimizer believes about $S_{12,12}$ — the design is complete at the single-qubit spectra, as on the anchored model. For the *idle*, the operation with genuine two-qubit freedom, the single-qubit-only rung pays: a factor $1.26\times$ at 640τ , $1.42\times$ at 1280τ , and $1.70\times$ at the deepest idle ($10240\tau \approx 3T_2^*$), landing on the full winner only where the search happens to stagger anyway. Restoring the ZZ self-spectrum (diag-3) recovers the six-spectrum design *exactly*, at every gate time — a designer who cannot see the coupler line parks ZZ filter weight on it, and $S_{12,12}$ is precisely what a single-qubit campaign cannot measure. Beyond diag-3 the ladder is flat by theorem (the cross-spectra are first-order invisible to average fidelity; see below).

SPAM-biased spectra. Handing the optimizer the raw arm’s biased reconstruction yields the *identical winner at the identical true infidelity* for both gates — the same NT(29,29) entangling design at 3.94×10^{-4} and the same idle winner at 4.36×10^{-4} as the reference and mitigated arms (Fig. 3, right column). On this landscape even the predictions agree across arms (within 0.5%, each sitting $\sim 1.5\times$ above its gate’s true value — the uniform conservatism discussed in Sec. 4): the raw arm’s corruption is concentrated on line heights and quiet ZZ teeth, frequencies where the *winning* filters carry little weight. That is landscape-dependent luck, not a property of unmitigated data — on the anchored model the raw arm mispredicted where the reference predicted to 0.7% — and it leaves SPAM’s measured cost here exactly where Fig. 2 shows it: in the spectroscopy. A device map drawn

from raw data mis-states the line heights by up to $\sim 2\times$ and the quiet ZZ floor by $\sim 9\times$; mitigation repairs all of it for two orders of magnitude less than the cost of being wrong about a trap’s depth.

What full, unbiased characterization buys is everything around the design:

- **The error budget.** Undecoupled, the two-qubit channels ($S_{12,12}$ plus cross-spectra) are a rounding error of free evolution (they contribute -3% at $T_G = 320\tau$: the anti-phase exchange coupling slightly *protects* the bare system). Under the best decoupling they carry **6.7%** of the entangling gate’s residual and **27.8%** of the deep idle’s (2560τ): decoupling suppresses the single-qubit channels so effectively that the two-qubit channels are promoted to over a quarter of what remains. Channel-resolved error budgeting of any decoupled gate *requires* two-qubit QNS.
- **The prediction.** The predictions on this landscape are uniformly conservative ($\sim 1.5\times$ above truth: the reconstruction reads the quiet band slightly high and treats the flagged DC points pessimistically), and *certified*: every uncertainty band in Sec. 4 exists only because the bars in Table 2 are unbiased with stated coverage. With biased central values the same bands would be confidently wrong.
- **The spectroscopy itself.** The raw arm’s spectra are relatively corrupted by up to $\sim 2\times$ on the line heights that set every trap’s depth and $\sim 9\times$ on the quiet ZZ teeth that set the ZZ floor; mitigation recovers the reference to 0.02σ .

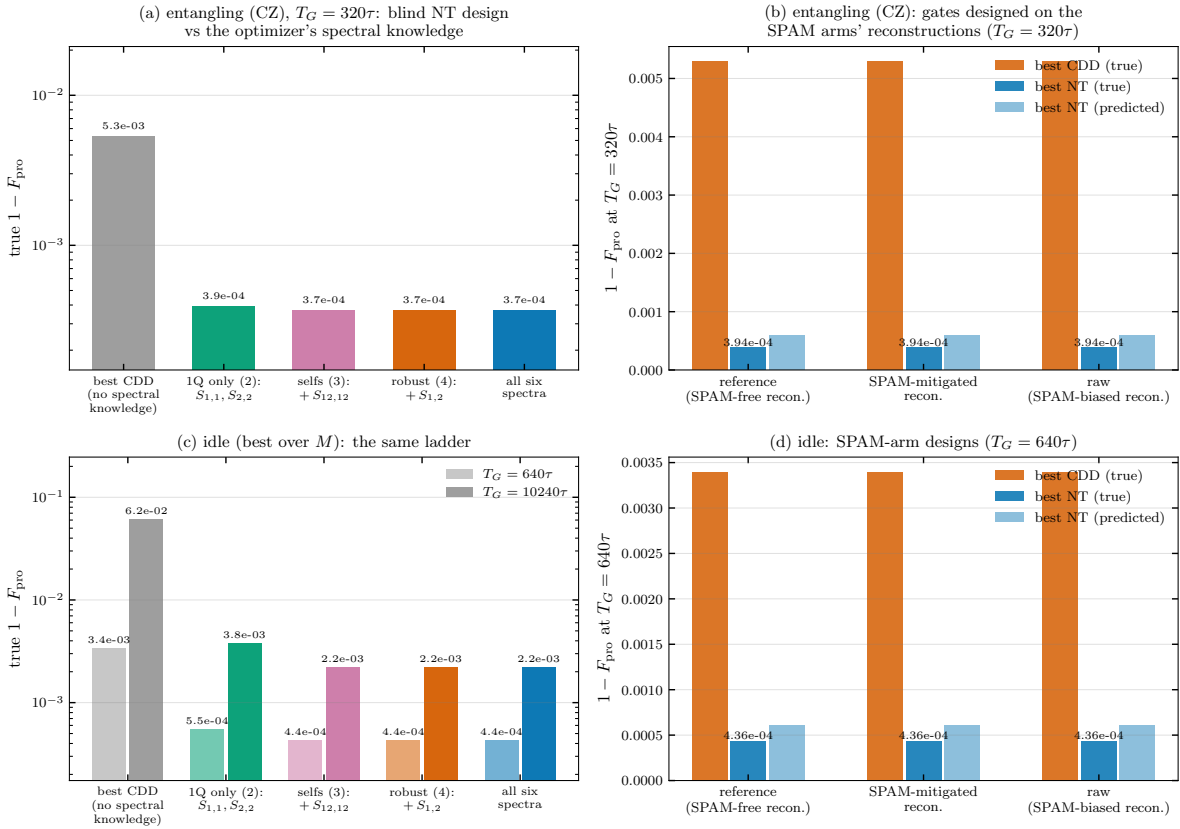


Figure 3: Both experiments for both gates: entangling (top row) and idle (bottom row, best over M ; light/dark bars: $T_G = 640\tau$ and 2560τ). Left column: true infidelity of the blind NT winner as the optimizer’s spectral knowledge grows from the CDD library’s none, through single-qubit-only, the three selfs, and the robust protocol’s four, to all six spectra. The CZ ladder is flat — its trains barely touch the two-qubit structure; the idle ladder is not — the coupler line is invisible to a single-qubit campaign, and the designs that cannot see it pay. Right column: gates designed on the SPAM arms’ reconstructions; true infidelity (solid) and the designer’s own prediction (light). The raw arm’s bias surfaces in the prediction, not the design.

What the cross-spectra are for. The off-diagonal spectra ($S_{1,2}, S_{1,12}, S_{2,12}$) barely move either ladder above — and that is a theorem, not a landscape accident: in every PTM element the cross-terms enter through $\pm$ filter

combinations whose first-order contributions cancel pairwise in the *average* fidelity, surviving only at $O(\chi^2)$. Average fidelity is the wrong meter for them. Their meter is element-resolved: *which* coherence survives. The idle’s defining task makes this concrete (Fig. 4): hold half of a Bell pair, $\Phi^+ = (|00\rangle + |11\rangle)/\sqrt{2}$, through the idle. The Bell coherences are exactly ZZ -immune ($|00\rangle$ and $|11\rangle$ share the $Z_1 Z_2$ eigenvalue), so only $S_{1,1}, S_{2,2}$ and $\text{Re } S_{1,2}$ enter: Φ^+ decays through the $S_{1,1} + S_{2,2} + 2\text{Re } S_{1,2}$ combination under in-phase decoupling ($y_2 = +y_1$: simultaneous pulses, the hardware default) and through $S_{1,1} + S_{2,2} - 2\text{Re } S_{1,2}$ under anti-phase decoupling ($y_2 = -y_1$: the same train with one qubit’s idle bracketed by frame flips). With the carrier common-mode at $c_{\text{QS}} = 0.85$, the bare physics is an $11\times$ split: in *free* evolution, where nothing suppresses the carrier, Ψ^+ outlives Φ^+ by the full $(1+c)/(1-c)$ factor (5.0×10^{-3} vs 5.6×10^{-2} at 320τ) — a decoherence-free-subspace structure that no single-qubit measurement can reveal. Under informed decoupling the carrier is largely suppressed and the choice still pays: the two implementations of the *same pulse train* — identical single-qubit marginals, average fidelities equal to *seven digits* (computed ratio 1.000 at every gate time), indistinguishable to any single-qubit QNS campaign — store the pair $1.2\times$ to $3.1\times$ apart, the split growing with storage time as the carrier’s share of the decoupled residual grows. Which implementation protects the pair is decided by the *sign* of the measured low-frequency $\text{Re } S_{1,2}$ (here > 0 , so anti-phase wins; on a device with anti-correlated drift the assignment flips); the magnitude, measured by the comb at 5σ -plus, prices the payoff. The blind reconstruction predicts the right parity ordering at every gate time, and a record/replay Monte Carlo through the synthesized noise reproduces the formula values to 0.1–1.3%. The blind average-fidelity NT winner sits between the two parities — correctly, since average fidelity cannot see the choice. This is the cross-spectra story in one figure: invisible to average fidelity, decisive for entanglement storage, and only two-qubit QNS can measure it.

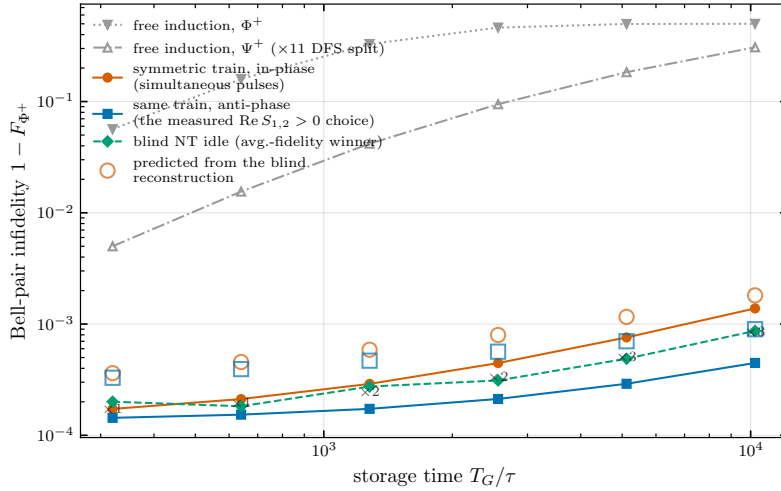


Figure 4: Holding a Bell pair through the idle. Free induction realizes the bare $(1+c)/(1-c) \approx 11\times$ split between Φ^+ (grey, filled) and Ψ^+ (grey, open): correlated-noise physics invisible to single-qubit data. Under decoupling, the same symmetric train implemented in-phase (simultaneous pulses, orange) and anti-phase (frame-flipped, blue) — implementations with identical average fidelity — still split by up to $3.1\times$, growing with storage time; open markers are the values predicted from the blind reconstruction, which picks the right parity everywhere. The blind average-fidelity NT idle (green) sits between the parities, as it must.

4 The gates: margins for both operations

Figure 5 summarizes the gate results on the 128k reconstruction, all under the 8τ pulse-spacing scenario and the blind protocol: select on the characterized spectra, report the true infidelity. Both June 12 requirements hold at once. *Realistic infidelities*: the tailored entangling gate holds at $3.5\text{--}9.5 \times 10^{-4}$ across the full $T_G = 40\text{--}2560\tau$ sweep — at 2560τ , where free evolution is at 45% error and the best library sequence at 4.2×10^{-2} , the tailored gate sits at 9.5×10^{-4} ; tailored idling holds at the low- 10^{-3} level out to $10240\tau \approx 3T_2^*$. *A dramatic, robust margin*: the entangling margin grows monotonically, $1.6 \rightarrow 3.1 \rightarrow 7.5 \rightarrow 14.3 \rightarrow 24.2 \rightarrow 35.5 \rightarrow 44.2\times$ across $T_G = 40\text{--}2560\tau$, and the idle margin $4.1 \rightarrow 27.7\times$ over $320\text{--}10240\tau$, for the structural reason of Sec. 1: each

doubling of the gate time slides the CDD library’s passbands down onto the next harmonic of the defect line, while the tailored design keeps using the measured quiet band.

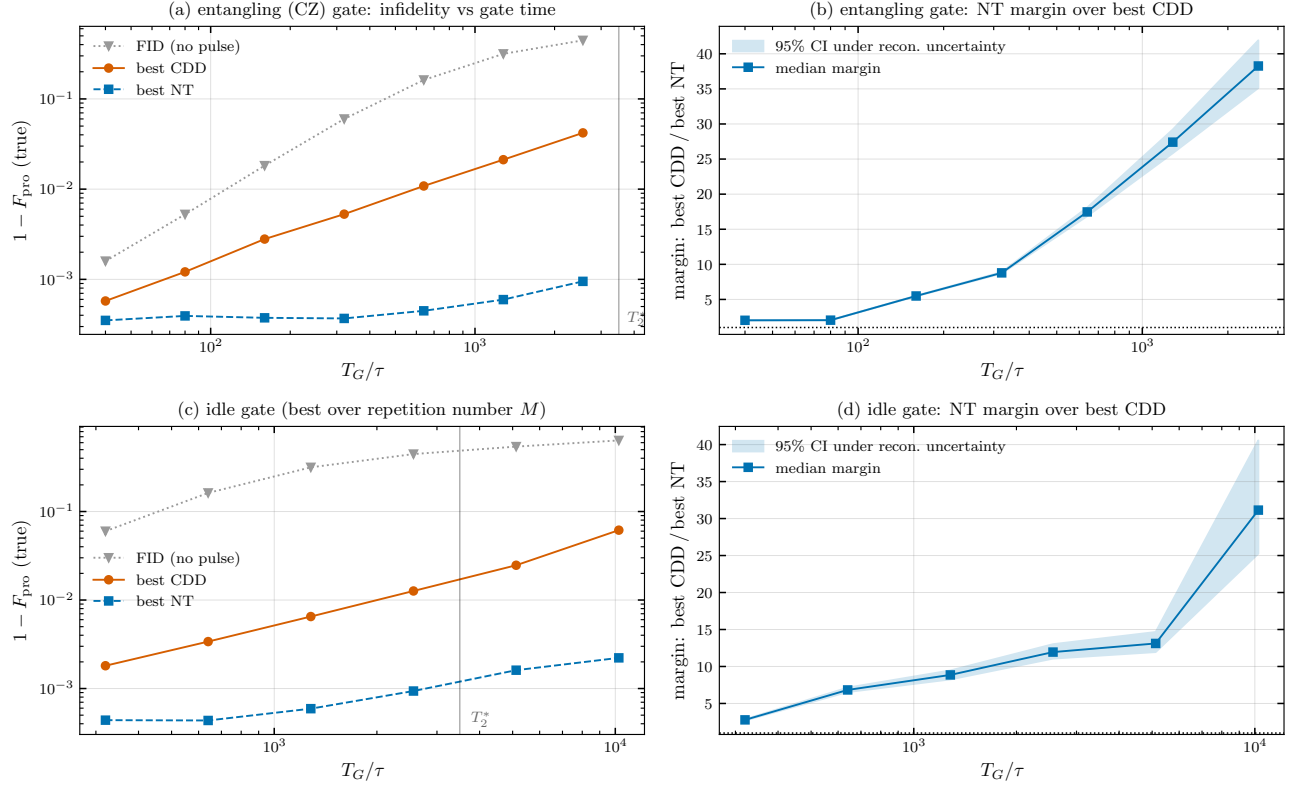


Figure 5: Noise-tailored gates on the featured model, all design blind, all scoring on truth, pulse-spacing constraint 8τ throughout. (a,c): true process infidelity vs. gate time for the entangling and idle gates: free induction (grey), best CDD (blind-selected, orange), best NT (blind-optimized, blue); the vertical line marks $T_2^* = 3500\tau$ (the idle is optimized over the repetition number $M \leq 128$ at each gate time). (b,d): the corresponding NT-over-CDD margin with the 95% band obtained by propagating the reconstruction’s quoted statistical and systematic uncertainties (systematics drawn coherently per channel, tail fits redone per draw) through both fixed winner sequences; the band is on the *ratio*, so common-spectrum fluctuations cancel.

The margins, with their 95% confidence intervals under reconstruction uncertainty:

- **Entangling (CZ):** $1.6 \times [2.01, 2.04]$ at 40τ ; $3.1 \times [2.02, 2.07]$ at 80τ ; $7.5 \times [5.44, 5.53]$ at 160τ ; $14.3 \times [8.69, 8.90]$ at 320τ ; $24.2 \times [16.9, 18.1]$ at 640τ ; $35.5 \times [25.8, 29.2]$ at 1280τ ; and $44.2 \times [35.1, 41.9]$ at $2560\tau \approx 0.8 T_2^*$.
- **Idle:** $4.1 \times [2.70, 2.86]$ at 320τ ; $7.8 \times [6.54, 7.14]$ at 640τ ; $11.0 \times [8.25, 9.46]$ at 1280τ ; $13.5 \times [11.1, 13.0]$ at 2560τ ; $15.3 \times [12.0, 14.7]$ at 5120τ ; and $27.7 \times [25.3, 40.6]$ at $10240\tau \approx 3 T_2^*$.

In each entry the central value is the *true* margin and the bracket is the 95% interval of the *predicted* margin under reconstruction uncertainty. From 160τ on, the bands track the true margins and sit conservatively below them: the reconstruction’s slightly-above-truth reading of the quiet band makes every predicted NT infidelity an upper estimate — the pipeline does not flatter its own gates. At the two shortest gate times, where best CDD and best NT sit within $2\text{--}3\times$ of each other at the few- 10^{-4} floor, the predicted ratio is least accurate (true $1.6 \times / 3.1 \times$ vs. predicted $\sim 2\times$); the quoted true values are the honest ones there.

Where this leaves the two-scenario story: the anchored model says that on a quiet, monotonic, well-behaved device, generic DD is nearly optimal and tailoring buys a reproducible $1.5\text{--}1.7\times$. The featured model says that the moment a device carries resolvable structure — defect harmonics, coupler resonances, common-mode drift, the things purification demonstrably does not remove — the same characterization pipeline, run once at a budget a real experiment can afford, separates 10^{-2} -class from few- 10^{-4} -class two-qubit gates; that the parts of the

operation	FID	best CDD	best NT	margin
idling 640τ	$1.62\text{e-}1$	$3.39\text{e-}3$	$4.36\text{e-}4$	$7.8\times$
idling 2560τ	$4.46\text{e-}1$	$1.27\text{e-}2$	$9.37\text{e-}4$	$13.5\times$
entangling 320τ	$5.99\text{e-}2$	$5.28\text{e-}3$	$3.69\text{e-}4$	$14.3\times$

Table 3: Sweet-spot numbers on the featured model (true process infidelities, blind designs, 8τ pulse-spacing scenario).

reconstruction the average fidelity does not consume are exactly the parts that decide channel-resolved budgets and entanglement storage; and that no shortcut (a single-qubit campaign, unmitigated SPAM) delivers the same gates with the same certified predictions. Which device one has is an empirical question — and answering it is itself the reconstruction.

All optimization blind; true infidelities evaluated exactly on the analytic model; characterization at 128k shots per estimator, SPAM arms at 64k on identical noise; pulse budgets separation-limited under the 8τ spacing scenario for both the NT search and the CDD library.